

Matthew Fong

Portfolio: <https://matthewfong3.github.io/>

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Projects:

Deserted Game (Unreal Engine 5)

- Implemented Enhanced Player Input Subsystem for player controls, such as movement, crouching, jumping, shooting, reloading, aim down sight (ADS) and smooth transition logic between player states
- Developed weapon struct system allowing implementation of various different types of weapons
- Implemented different weapon gun fire modes (automatic fire, semi-automatic fire, burst fire) on different weapons, allowing the player to switch a weapon's fire mode
- Scripted a time trial minigame feature in the shooting range level
- Implemented a weapon buy system similar to Call of Duty Zombies mode, where the player can spend points to purchase weapons around the map or replenish ammo

Interlinked (Unreal Engine 5)

- Designed the code architecture of the player's 6 ability spells
- Programmed 2 of the 6 player's ability spells with UE5's Gameplay Ability System (GAS)
- Designed and implemented 2 of 3 enemy AI's behavior trees
- Implemented procedural walk animations with UE5's control rig and locomotor
- Implemented animation blueprint and animation montages for Torture Nexus, an AI enemy

TPS Multiplayer Demo (Unreal Engine 5)

- Programmed player controls with the Enhanced Input Subsystem, such as movement, sprinting, jumping, crouching, shooting, aiming down sight (ADS), and reloading with C++
- Programmed replication of line tracing when shooting, applying damage, reloading, sound effects and particle effects from server across all clients through C++
- Programmed the logic behind updating weapon ammunition when reloading
- Programmed the animation blueprint for the third person character
- Designed and whiteboxed a simple level to demo the multiplayer prototype

Configura (Godot)

- Refactored existing codebase to be more performant
- Programmed the extraction of the base model meshes into a single scene and accessory meshes into their own respective scenes
- Programmed the feature to dynamically loading new mesh assets and unloading previously selected assets in the character creator
- Created documentation for developers to reference descriptions of variables, methods, and classes of all scripts

Skills:

Programming Languages: C++, C#, GDScript, JavaScript

Software: Unreal Engine 5, Blender, Godot, Unity, Visual Studio, VS Code, Autodesk Maya

Concepts: Object-Oriented Programming, Agile Development, Version Control, Data Structures & Algorithms, UI/UX Design, RESTful-API

Tools/Frameworks: Github, Trello, Jira, HTML5, CSS3, ES6, NodeJS, ReactJS, Websockets (socket.io), ExpressJS, MVC framework, Sass, Flexbox

Education:

Rochester Institute of Technology

Master of Science in Game Design and Development

Rochester, NY

Expected May 2027

Relevant Coursework: Advanced Topics in Game Dev (Unreal Engine 5), AI for Gameplay, Production Studio, Gameplay and Prototyping, Game Development Processes, Game Design

Overall GPA: 4.00/4.00

Bachelor of Science in Game Design and Development, Immersion in Music

May 2018

Relevant Coursework: Game Design & Development, Game Graphics Programming, Data Structures & Algorithms, Game Dev & Algorithm Problem Solving, Level Design, 3D Animation & Asset Production

Honors and Awards: Summa Cum Laude, Dean's List Fall 2014 - Spring 2018

Work Experience:

Magic Spell Studios

Traver Creative Technologist Founders Program Developer

Rochester, NY

May-August 2026

- Developing Configura, an open-source character creation tool built in Godot
- Maintaining and expanding Configura's core functionality, including its character creator add-on
- Developing new features
- Improving existing systems
- Collaborating with 3D artists to integrate character assets
- Promoting the project through online communities

The Langham Fifth Avenue, New York

Guest Service Ambassador

New York, NY

May 2022-April 2024

- Collaborated with supervisors and team members in brainstorming new ideas to improve workflow efficiency
- Flexibility in assisting supervisor and team members with various tasks as necessary during peak business hours
- Listened to guest complaints, diffusing tense situations and implementing solutions to ensure maximum guest satisfaction
- Collaborated with other department managers to ensure upcoming deadlines were met
- Clear and prompt communication between different departments within the company, regarding special guest needs, requests, inquiries
- Regularly following up with guests to gain feedback and ensure their stay and service provided was nothing less of perfection